

THRISTHA GURAJALA

AI/ML Engineer

San Francisco, CA Open for relocation to Seattle, WA | Plano, TX | Chicago, IL | New York, NY

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SUMMARY

AI/ML Engineer with 5+ years of experience designing, deploying, and optimizing large-scale LLM and multimodal AI systems across cloud and edge environments. Proven success in reducing inference latency, improving model accuracy, and boosting user engagement through production-grade RAG pipelines, reinforcement learning, and multi-agent orchestration. Skilled in PyTorch, Llama, Sonar LLM, FAISS, AWS, and Kubernetes, with strong knowledge of GCP Vertex AI and Azure ML fundamentals. Experienced in implementing Responsible AI, PCI-DSS compliance, and automated safety guardrails (LangSmith, DSPy, Guardrails.ai). Adept at translating complex AI solutions into secure, scalable products and mentoring cross-functional teams to deliver measurable business impact.

PROFESSIONAL EXPERIENCE

Perplexity.ai (Comet Browser)	May 2025 - Present
AI/ML Engineer	San Francisco, CA
- Reduced average inference latency by 42% through hybrid orchestration of on-device quantized models and AWS GPU inference clusters using speculative decoding, cutting end-user wait time from 2.6 s → 1.5 s. - Increased AI task completion accuracy by 31% via reinforcement-driven retrieval-augmented generation (RAG) pipeline and fine-tuned Sonar LLMs with context-aware embeddings. - Improved user engagement by 27% MoM by launching personalization and memory layers using encrypted local storage + vector similarity ranking (FAISS on AWS EKS), driving measurable user retention lift. - Designed and deployed distributed inference architecture on AWS EKS leveraging PyTorch 2.x, Triton Inference Server, and Ray Serve, with autoscaling and GPU node pool segregation for 10K+ concurrent sessions. - Engineered real-time media processing pipelines with Kafka + Flink, FFmpeg, and Whisper STT to enable multimodal summarization (video/audio/text) inside the Comet Browser assistant. - Implemented secure browser-side AI agents using TypeScript + WebGPU + WASM for low-latency summarization and form-action inference directly within Chromium render processes. - Built and optimized retrieval layer using FAISS + Elasticsearch for 20 B+ documents, enabling sub-200 ms vector lookup times and citation-aware contextual grounding. - Automated ML ops pipelines via AWS SageMaker, Kubernetes, and GitHub Actions, integrating model registry, CI/CD, and A/B testing across production Sonar LLMs. - Integrated AdTech and payment microservices (Stripe API, AWS Lambda, DynamoDB) with revenue-share logic for Comet Plus subscriptions and publisher monetization dashboards. - Deployed advanced observability stack (Prometheus, Grafana, OpenTelemetry, Sentry) with real-time drift detection and prompt-injection monitoring, improving incident resolution speed by ~45%.	
Meta	May 2024 – April 2025
Software Engineer	San Francisco, CA
- Optimized Llama model inference pipelines, reducing GPU latency by 28% and improving throughput for large-scale deployments across Meta AI’s production clusters on AWS. - Enhanced personalized ad recommendation accuracy by 19% using hybrid embedding models integrated with Meta’s Andromeda retrieval engine and Llama-based contextual embeddings. - Cut multimodal media processing costs by 24% by implementing quantization-aware training and FP8 precision across distributed AWS GPU nodes, improving both speed and energy efficiency. - Engineered end-to-end LLM pipelines for Llama 3.x models using PyTorch, TensorRT-LLM, and Ray, enabling distributed fine-tuning and inference scaling to billions of parameters. - Developed RAG (Retrieval-Augmented Generation) workflows leveraging Faiss, Redis, and AWS S3 for low-latency document retrieval and context augmentation across Meta AI’s Llama Stack APIs. - Built real-time feature streaming and user modeling layers with Apache Flink, Kafka, and PyTorch Serve, powering personalization and ad ranking in Meta’s production environment. - Integrated vision-language fusion modules for multimodal Llama 3.2 models using CLIP-style encoders, AITemplate optimizations, and adaptive media ingestion pipelines. - Deployed and monitored AI services on AWS EKS and EC2 GPU clusters, implementing CI/CD with Terraform, Docker, and Kubernetes for reproducible LLM deployments. - Contributed to open-source Llama Stack APIs (Inference, Agents, Safety, and Evaluation layers), improving developer usability, telemetry integration, and cross-platform support (cloud + on-device).	

- Assisted in deploying internal AI/ML accelerators on AWS, contributing to a 25% improvement in test automation throughput across multiple enterprise applications.
- Supported real-time personalization and recommendation modules, helping reduce latency of decision-making pipelines by 30%, enhancing platform efficiency for internal users.
- Developed analytics dashboards for monitoring ML model performance, enabling early detection of drift and improving predictive accuracy by 20%.
- Assisted in integrating NLP and computer vision models using PyTorch and TensorFlow into microservices architecture on AWS, supporting scalable AI services.
- Supported building data pipelines with Apache Spark, Kafka, and AWS S3, ensuring high-throughput ingestion of structured and unstructured datasets for ML workflows.
- Developed real-time model inference services with Flask/FastAPI and Kubernetes on AWS, enabling low-latency processing for automation and media pipelines.
- Analyzed feature sets, performed data preprocessing, and implemented ML experiments using scikit-learn and MLflow, contributing to reproducible and version-controlled AI workflows.

TECHNICAL SKILLS

Languages & Frameworks: Python (FastAPI, Flask), TypeScript, JavaScript (ES6+), C++ (basic), Java (basic)
AI/LLM Development: Llama 3.x, Sonar LLM, PyTorch 2.x, Hugging Face Transformers, DeepSpeed, ONNX, FSDP, bitsandbytes, PEFT, LoRA/QLoRA, Prompt Engineering (CoT, Few-Shot, RLAI), vLLM
RAG & Agentic AI Systems: LangChain, FAISS, Elasticsearch, Redis, Pinecone, Retrieval-Augmented Generation (RAG), LangGraph, CrewAI, Context-Aware Embeddings, Citation-Aware Grounding, Guardrails.ai
Data Science & ML: scikit-learn, Pandas, NumPy, TensorFlow, XGBoost, Spark, Statistical Modeling, Feature Engineering, Model Evaluation Metrics, A/B Testing, Responsible AI
Cloud Platforms & MLOps: AWS (EKS, Lambda, SageMaker, EC2, S3), Kubernetes, Docker, Terraform, Ray Serve, TorchServe, Triton Inference Server, MLflow, CI/CD Pipelines, Distributed Systems, Autoscaling & Cluster Management - familiar with GCP Vertex AI, Azure ML & Azure OpenAI fundamentals
Databases & Caching: DynamoDB, PostgreSQL, MongoDB, Redis, Query Optimization, Relational & Non-Relational Databases
Streaming & Real-Time Processing: Kafka, Apache Flink, Low-Latency Pipelines, Multimodal Media Processing (Audio/Video/Text), FFmpeg, Whisper STT
Monitoring & Observability: Prometheus, Grafana, OpenTelemetry, Sentry, Drift Detection, LangSmith, DSPy, Prompt Injection & Hallucination Monitoring
Web & Edge Computing: WebGPU, WASM, Browser-Side AI Agents, Chromium Extensions
AdTech & Payments Integration: Stripe API, Revenue-Share Logic, Publisher Dashboards, PCI-DSS, HIPAA & SOX Compliance
Soft Skills: Problem Solving, Strategic Thinking, Collaboration, Rapid Learning, Decision-Making, Technical Communication, Mentoring, Adaptability

EDUCATION

University of Tampa
Masters in cybersecurity